14 ENERGY TRANSFERS

Whenever anything happens, energy is transferred from one store to another – indeed, without energy things simply can't happen! In this chapter, you will learn that energy can be transferred to many different stores, including sound, light, movement, heat and potential energy. You will also find out that, although energy is never destroyed, in every energy transfer some energy is transferred to the surroundings, often as heat.





▲ Figure 14.1 Energy is stored as sound, movement and light, for example.

LEARNING OBJECTIVES

- Describe energy transfers involving energy stores:
  - energy stores: chemical, kinetic, gravitational, elastic, thermal, magnetic, electrostatic, nuclear
  - energy transfers: mechanically, electrically, by heating, by radiation (light and sound)
- Use the principle of conservation of energy
- Know and use the relationship between efficiency, useful energy output and total energy output:

$$\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy output}} \times 100\%$$

- Describe a variety of everyday and scientific devices and situations, explaining the transfer of the input energy in terms of the above relationship, including their representation by Sankey diagrams

UNITS

In this section you will need to use kilogram (kg) as the unit of mass, joule (J) as the unit of energy, metre (m) as the unit of length, metre per second (m/s) as the unit of speed and velocity, metre per second squared (m/s^2) as the unit of acceleration, newton (N) as the unit of force, second (s) as the unit of time and watt (W) as the unit of power. You have used all these units before.

For things to happen we need energy! Energy is used to produce sound. Energy is used to transport people and goods from place to place, whether it is by train, boat or plane or on the backs of animals or even by bicycles. Energy is needed to lift objects, make machinery work and run all the electrical and electronic equipment we have in our modern world. Energy is needed to make light and heat. The demand for energy increases every day because the world's population is increasing. People consume energy in the form of food and need energy for the basics of life, like warmth and light. As people become wealthier they demand much more than the basics, so the need for energy grows!

STORES OF ENERGY

Energy is found in many different stores:

- We get our energy from the food we eat. Food provides stored chemical energy that we can burn to transfer it to other types of energy.
- We use the energy from food to generate **thermal** energy (heat energy) to help to keep us warm.

EXTENSION WORK

Sound energy can be used to break up small stones that can form inside a person's body, without the need for an operation. Sound energy is also used in medicine to examine the inside of the body, as an alternative to x-rays.

- Our muscles transfer the chemical energy to movement energy (kinetic energy).
- Some of this movement energy is transferred as we speak to sound energy.
- We need heat for our homes, schools and workplaces. We also need to transfer energy to light for our buildings, vehicles and roads.
- Most of the energy needed for these purposes is transferred from electrical energy. We shall see later (see Chapter 17) that electrical energy can be transferred from other stores of energy, like chemical or nuclear energy.
- Some electrical energy is produced from the gravitational potential energy in water kept in reservoirs in mountainous areas. We can also use the energy in the hot core (centre) of the Earth. We see the evidence of this huge supply of heat in volcanoes and thermal springs.
- Energy from the heat underground is called geothermal energy.
- Energy can also be stored in springs as elastic potential energy. This type of stored energy is used in things like clocks and toys.

The main resource of energy for the Earth is our Sun. This provides us with heat, light and other stores of energy.

ENERGY TRANSFERS

For energy to be useful, we need to be able to transfer it from one store into whichever store we require. Unfortunately, when we try to do this there is usually some energy transferred to unwanted stores. We often refer to these unwanted stores as 'wasted' energy because it is not being used for a useful purpose.

Here are some examples of wasted energy:

- An electric heater may be used to heat water in a house. The hot water will be stored in a tank (container) for use when required. Although the tank may be well insulated, some energy will be transferred from the water and some will be used to heat up the copper that the tank is made from. Both processes mean that some of the energy that is transferred from electricity to heat is wasted because it is not doing what we want it to do – it is not making the water hot.
- When we fill a car with petrol, the main purpose is to transfer the chemical energy stored in the fuel to movement energy, with a small amount doing other things like providing electrical energy for the lights or radio. But the process of transferring the energy is not perfectly efficient, as not all of the energy is used to do what we want. (A formal definition of efficiency is given on page 137.) A considerable amount of the energy supplied by the fuel is transferred to heat, most of which is transferred to the surroundings. Some of the energy is transferred to sound, which can be unpleasant for people both inside and outside the car. A lot of energy is transferred to overcome the various friction forces that oppose movement of parts within the car and the car's movement along the road. Friction causes energy transfers to occur too, usually producing unwanted heat energy.

Unwanted energy transfers reduce efficiency. This problem is the same whether the system is a small one, like a car, or a large system, like the nationwide electricity generation and distribution industry. We need to be aware of where our energy is transferring to if we are to find ways of using it well.

SOME ENERGY TRANSFERS

We have many ways of transferring energy from one store to another.



▲ Figure 14.2 Energy is transferred from one store to another, and to another, and so on.

In Figure 14.2, stored chemical energy in the food is needed to help our bodies make a range of other chemicals. Some of these – like carbohydrates – are used to produce heat to maintain our body temperature and energy for movement through muscle activity. Having eaten a meal, the cyclist in Figure 14.2 is transferring the chemical energy stored in his body to movement energy. The movement is initially in the cyclist's legs and is then transferred to the machine (the bicycle). The cyclist is also transferring additional heat energy, which is then transferred to the surroundings. Friction in various parts of the bicycle will also result in energy being transferred from movement to heat and sound.

The dynamo or generator fitted to the bicycle's wheel transfers some of the movement energy of the wheel to electrical energy. The lamp then transfers the electrical energy to light and heat. As the electrons flow in the dynamo–lamp circuit, heat will be produced in the conducting wires too.

Examples of other energy transfers are given in the questions at the end of the chapter.

CONSERVATION OF ENERGY

The principle of conservation of energy is a very important rule. It states that:

Energy is not created or destroyed in any process.

(It is just transferred from one store to another.)

We often hear about the energy crisis: as our demand for more energy increases our reserves of energy in the form of fuels like oil and gas are rapidly being used up. The principle of conservation of energy makes it seem as if there is no real problem – that energy can never run out. We need to understand what the principle really means.

Physicists believe that the amount of energy in the Universe is constant – energy can be transferred from one store to another but there is never any more or any less of it. This means we cannot use energy up. However, if we consider our little piece of the Universe, the problem becomes more obvious. As we make energy do useful things – for example, move a car – some of it will be transferred to heat. Some of this heat energy will be radiated away from the Earth and transferred (not destroyed or used up) into space. This means the

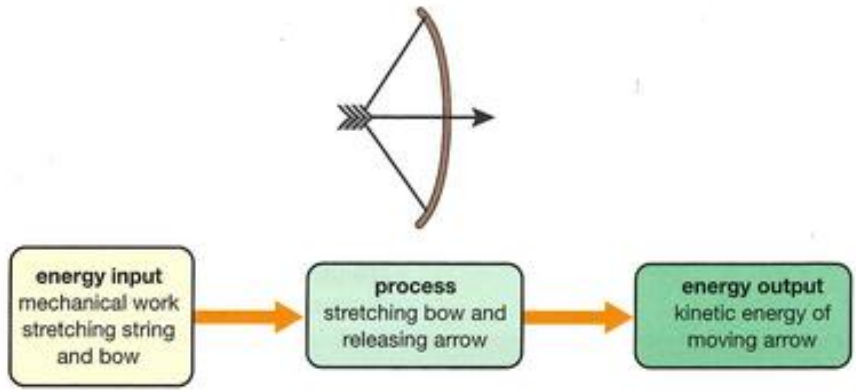
EXTENSION WORK

Even though the amount of energy in the Universe is constant, it is becoming more spread out and so less available for use. Some scientists think that all the energy in the Universe will eventually be transferred to heat and that everywhere in the Universe will end up at the same very low temperature. This possible 'end of the world' is sometimes referred to by the dramatic name of 'heat death'. If the Universe does end up this way, it is not expected to do so for some time yet, so carry on working for those exams!

energy is not available for us to use any more. This is just like a badly insulated house; if heat energy escapes, it is transferred away from the system we call our home, and is no longer available to keep us warm.

SANKEY DIAGRAMS

We use different ways to show how energy is transferred. Energy transfer diagrams show the energy input (contribution), the energy transfer process and the energy output (production). The system may be a very simple one with just one main energy transfer process taking place. An example of a simple system with its energy transfer diagram is shown in Figure 14.3.

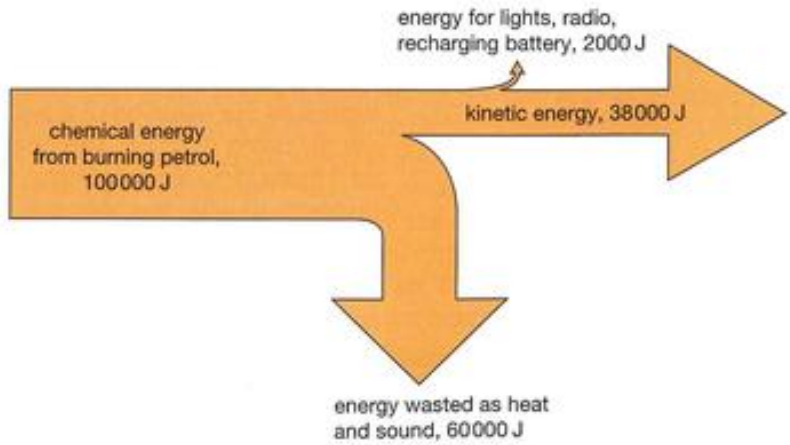


▲ Figure 14.3 Energy transfer diagram for an arrow being fired from a bow

Sankey diagrams are a simpler and clearer way of showing what happens to an energy input into a system. The energy flow is shown by arrows whose width is proportional to the amount of energy involved. Wide arrows show large energy flows, narrow arrows show small energy flows.

Figure 14.4 shows a Sankey diagram for a complex system – the energy flow for a car. Chemical energy in the form of petrol is the input to the car. The energy outputs from the car are:

- electrical energy (from the alternator) to drive lights, radio and so on, to charge the battery (transferred to chemical energy) and allow the car to switch on
- movement (kinetic) energy from the car engine
- wasted energy as electrical heating in wiring and lamp filaments, as frictional heating in various parts of the engine and alternator, and as noise.



▲ Figure 14.4 Sankey diagram showing the energy flow in a typical car

Here the 100 000 J of chemical energy input might be shown by an arrow 20 mm wide, so the 60 000 J of energy wasted (60% of the input) would then be shown by an arrow that is 12 mm wide (60% of 20 mm). It is difficult to draw the 2% arrow for the energy output to scale; it is enough to show it as very small.

EFFICIENCY

Whenever we are considering energy transfers, we have to remember that a proportion of the energy input is wasted. Remember that wasted means transferred into stores other than the useful store required. We would like our energy transfer systems to be perfect with all the output energy being in the store that we want. For example, not all the output energy for an electric lamp is light (the useful energy output), some of the output energy is heat (not useful when what we want is light).

Real systems always have an unwanted energy output so can never have 100% efficiency.

The efficiency of an energy conversion system is defined as:

$$\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy output}} \times 100\%$$

HINT

If you calculate the efficiency of a system and get an answer bigger than 100% then you have put the numbers into the equation the wrong way round!

Efficiency does not have a unit because it is a ratio. Sometimes efficiency is shown as a **fraction** of the energy output that is in the wanted or useful store. In real energy transfers this fraction will always be less than 1 because some of the total output energy will be in an unwanted store.

EXAMPLE 1

A 60 W tungsten filament bulb uses 60 J of energy every second. It is 5% efficient. How much of the total energy output per second is useful light energy?

$$\text{efficiency} = \frac{\text{useful energy output}}{\text{total energy output}} \times 100\%$$

$$5\% = \frac{\text{useful energy output}}{60\ \text{J}} \times 100\%$$

$$\frac{5}{100} = \frac{\text{useful energy output}}{60\ \text{J}}$$

So useful (light) energy from bulb $= \frac{5 \times 60\ \text{J}}{100} = 3\ \text{J}$ each second.

CHAPTER QUESTIONS

More questions on the need for energy can be found at the end of Unit 4 on page 168.

SKILLS CREATIVITY

1 Describe the main energy transfers taking place in the following situations:

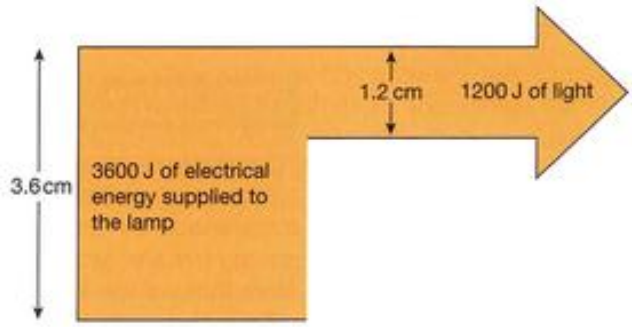
- a turning on a torch
- b lighting a candle
- c rubbing your hands to keep them warm
- d bouncing on a trampoline.

SKILLS INTERPRETATION, PROBLEM SOLVING

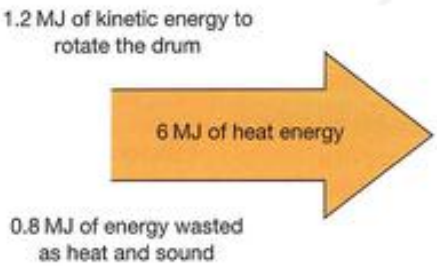
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2 Copy and complete the following Sankey diagrams. Remember that the width of the arrows must be proportional to the amount of energy involved. This has been done for you in part a.

a for an electric lamp



b In a typical wash in a washing machine 1.2 MJ is transferred to the kinetic energy of the rotating drum, 6 MJ of energy is transferred to heat the water and 0.8 MJ is wasted as heat transferred to the surroundings and sound.



SKILLS INTERPRETATION

3 a Draw a Sankey diagram for the following situation. An electric kettle is used to heat some water. 350 kJ of energy are used to heat the water, 10 kJ raise the temperature of the kettle and 40 kJ escape to heat the surroundings.

SKILLS PROBLEM SOLVING

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b Calculate the efficiency of the kettle.

4 A ball is dropped. It hits the ground with 10 J of kinetic energy and bounces with 4 J of kinetic energy.

6

a What happens to 6 J of the energy during the bounce?

SKILLS INTERPRETATION

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b Draw a Sankey diagram for the energy flow that takes place during the bounce.